

17-2c Other Examples of the Prisoners' Dilemma

We have seen how the prisoners' dilemma can be used to understand the problem facing oligopolies. The same logic applies to many other situations as well. Here we consider two examples in which self-interest prevents cooperation and leads to an inferior outcome for the parties involved.

Arms Races

In the decades after World War II, the world's two superpowers—the United States and the Soviet Union—were engaged in a prolonged competition over military power. This topic motivated some of the early work on game theory. The game theorists pointed out that an arms race is much like the prisoners' dilemma.

To see why, consider the decisions of the United States and the Soviet Union about whether to build new weapons or to disarm. Each country prefers to have more arms than the other because a larger arsenal would give it more influence in world affairs. But each country also prefers to live in a world safe from the other country's weapons.

Figure 3 shows the payoff matrix for this deadly game. If the Soviet Union chooses to arm, the United States is better off doing the same to prevent the loss of power. If the Soviet Union chooses to disarm, the United States is better off arming because doing so would make it more powerful. For each country, arming is a dominant strategy. Thus, each country chooses to continue the arms race, resulting in the inferior outcome with both countries at risk.

Figure 3

An Arms-Race Game

In this game between two countries, the safety and power of each country depend on both its decision whether to arm and the decision made by the other country.

		Decision of the United States (U.S.)	
		Arm	Disarm
Decision of the Soviet Union (USSR)	Arm	U.S. at risk USSR at risk	U.S. at risk and weak USSR safe and powerful
	Disarm	U.S. safe and powerful USSR at risk and weak	U.S. safe USSR safe

Throughout the Cold War era from about 1945 to 1991, the United States and the Soviet Union attempted to solve this problem through negotiation and agreements over arms control. The problems that the two countries faced were similar to those that oligopolists encounter in trying to maintain a cartel. Just as oligopolists argue over production levels, the United States and the Soviet Union argued over the amount of arms that each country would be allowed. And just as cartels have trouble enforcing production levels, the United States and the Soviet Union each feared that the other country would cheat on any agreement. In both arms races and oligopolies, the relentless logic of self-interest drives the participants toward the noncooperative outcome, which is worse for both parties.

Common Resources

In [Chapter 11](#) we saw that people tend to overuse common resources. One can view this problem as an example of the prisoners' dilemma.

Imagine that two oil companies—ExxonMobil and Chevron—own adjacent oil fields. Under the fields is a common pool of oil worth **\$12** million. Drilling a well to recover the oil costs **\$1** million. If each company drills one well, each will get half of the oil and earn a **\$5** million profit (**\$6** million in revenue minus **\$1** million in costs).

Because the pool of oil is a common resource, the companies will not use it efficiently. Suppose that either company could drill a second well. If one company has two of the three wells, that company gets two-thirds of the oil, which yields a profit of **\$6** million. The other company gets one-third of the oil, for a profit of **\$3** million. Yet if each company drills a second well, the two companies again split the oil. In this case, each bears the cost of a second well and therefore earns a profit of only **\$4** million.

[Figure 4](#) shows the game. Drilling two wells is a dominant strategy for each company. Once again, the self-interest of the two players leads them to an inferior outcome.

Figure 4

A Common-Resources Game

In this game between firms pumping oil from a common pool, the profit that each earns depends on both the number of wells it drills and the number of wells drilled by the other firm.

Chapter 17: Oligopoly: 17-2c Other Examples of the Prisoners' Dilemma

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